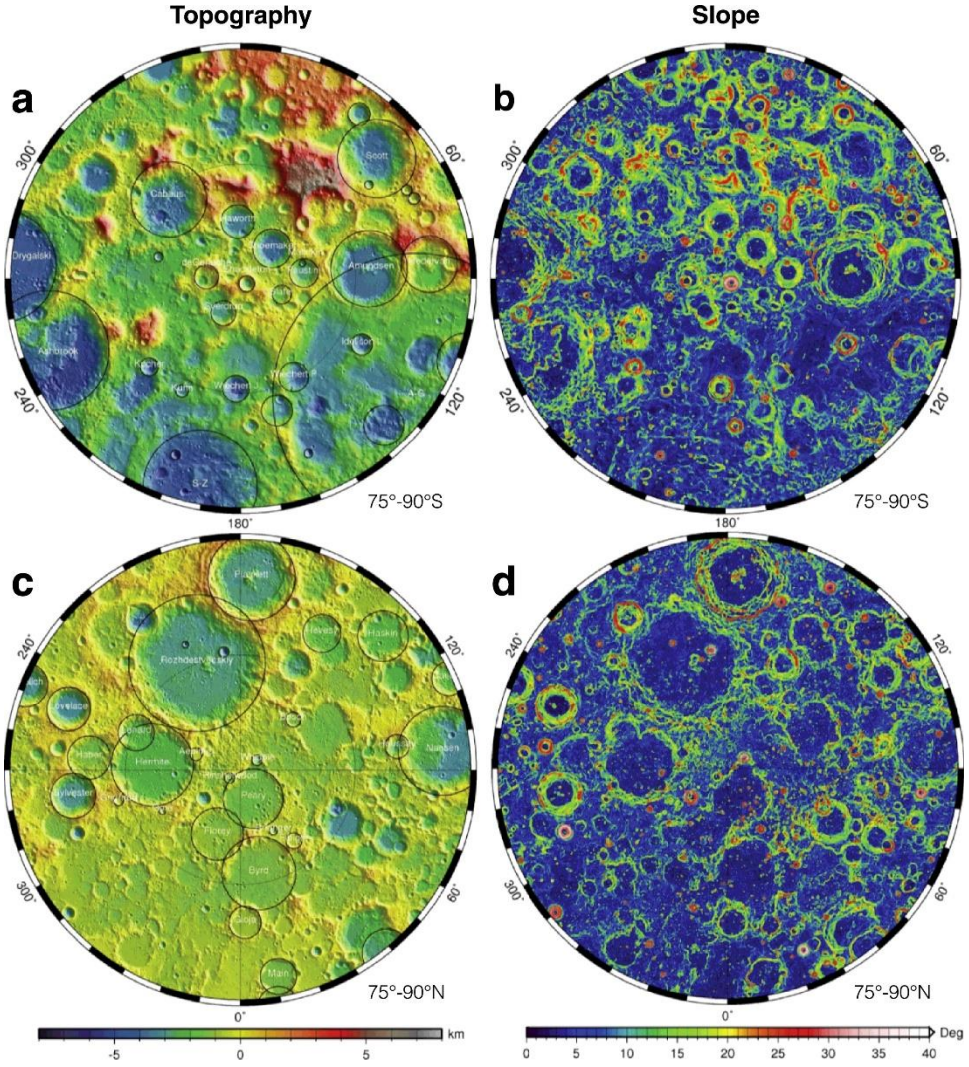



**Figure 3.4.1.3-4 LOLA topography (a, c) and surface slopes (b, d) at 100-m baseline for the lunar south (top) and north poles (bottom). Both maps extend from their respective pole to 75° latitude (from Smith et al., 2017).**

### 3.4.1.4 Rocks and Rock Size Distribution

Rocks are relatively rare on the lunar surface – the fractional area covered by rocks (diameter ~1m and above) is typically <1% (Hayne et al., 2017) and the average fractional area of the surface covered by rocks is 0.3% (e.g., Figure 3.4.1.4-1). However, rocks are not randomly distributed, and the fraction of the surface covered by rocks reaches high values (10-20%), albeit rarely (<0.01% of the surface). For this reason, hazards due to high rock abundances need to be locally assessed during detailed landing site characterization. The locations with the highest rock abundances are the ejecta of fresh craters, the interior of sizable and/or fresh craters, and on or beneath other steep slopes (e.g., rille walls). In addition to these locations, occasional rocks without an obvious source are observed; most of these are ejecta from distant impact craters.

Highlands and polar regions are generally less rocky than the maria, as smaller fresh craters are able to excavate bedrock in the maria (see Figure 3.4.1.4-3).

Rock size-distributions have been characterized from both *in situ* and orbital data. Where rocks are present, small rocks are much more common than large ones. It is common to fit observed rock size-frequency distributions with a power law (see Figure 3.4.1.4-2), where the cumulative number of rocks larger than size D is expressed as:

$$N_{>D} = CD^{\gamma}$$

Example results are given in Table 3.4.1.4-1, with all literature results normalized so that D is in m and N is the cumulative number density of rocks ≥D in an area of 100 $m^2$. These cumulative size-frequency distributions for rocks are based on lander observations primarily at decimeter scales, and are should be considered valid only for sizes <2m.

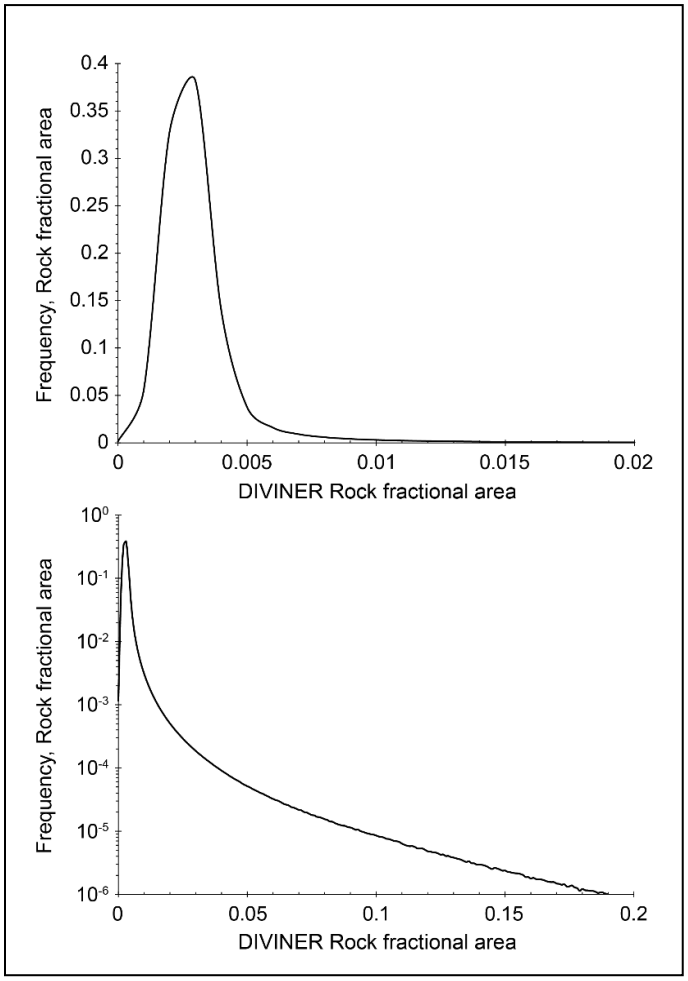



**Figure 3.4.1.4-1. Frequency of fractional area covered by rocks from Diviner, shown with a linear frequency axis (top) and logarithmic frequency axis (bottom). From PDS Diviner rock abundance grid, which is calculated at 236 m/px resolution.**

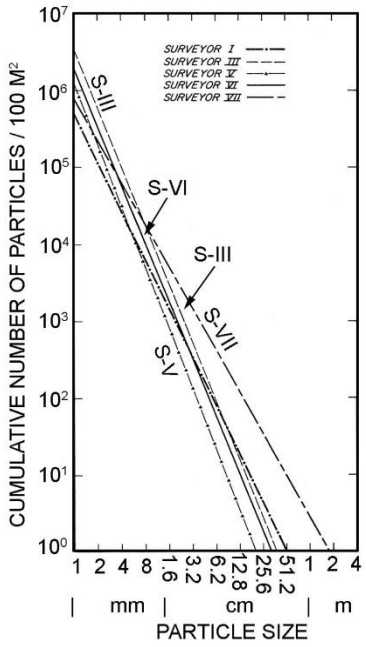



**Figure 3.4.1.4-2. Cumulative size-frequency distributions for rocks near the Surveyor spacecraft derived from surface images (from Shoemaker and Morris, 1970).**

**Table 3.4.1.4-1. Landing site analysis of rock abundance. All of these sites with the exception of Surveyor 7 – an extreme case on the rim of Tycho crater – have fractional rock area of <~1% in Diviner.**

| Landing Site | *C* (# ≥1 m / 100 m$^2$) | $\gamma$ | # ≥10 cm / 100 m$^2$ | # with *height* ≥ 1 m / 100 m$^2$ | Source Reference |
|---|---|---|---|---|---|
| Surveyor 1 | 0.234 | -2.11 | 30.1 | 0.08 | Shoemaker and Morris, 1970 |
| Surveyor III | 0.069 | -2.56 | 25 | 0.02 | Shoemaker and Morris, 1970 |
| Surveyor V | 0.014 | -2.65 | 6.26 | 0.00 | Shoemaker and Morris, 1970 |
| Surveyor VI | 0.058 | -2.51 | 18.2 | 0.02 | Shoemaker and Morris, 1970 |
| Surveyor VII | 2.739 | -1.82 | 181 | 1.08 | Shoemaker and Morris, 1970 |
| Apollo 16 Site 4 | 0.217 | n/a | 43.2 | 0.07 | Muehlberger et al., 1972 |
| Apollo 16 Site 5 | 0.384 | n/a | 76.6 | 0.12 | Muehlberger et al., 1972 |
| Apollo 16 Site 6 | 0.189 | n/a | 37.7 | 0.06 | Muehlberger et al., 1972 |
| Chang E'3 | 0.085 | n/a | 17 | 0.03 | Di et al., 2016 |

Diviner data suggests that the Surveyor 1 rock distribution represents a 91st percentile environment and could be used for initial design purposes when a specific landing site is unknown. With site selection and/or active hazard avoidance it is possible to access more than the 91st percent of the Moon with this envelope. For any specific mission, this data must be used in concert with LROC NAC high resolution imaging to establish site specific parameters. See Technical Notes for links.

The height of rocks (h), on average, is less than their visible diameter (D). Measurements at the Apollo and Lunokhod landing sites suggest an average ratio of h/D=0.60 with σ=0.28 (Demidov and Basilevsky, 2014). Rock shape and angularity of rocks on the Moon is variable and site specific. The predominant rock shape can range from mostly rounded to mostly angular at different locations on the lunar surface. An abundance of angular rocks is most common in areas where immature material has been recently excavated from depth, e.g., by a fresh impact crater.

**Technical Notes**

Knowledge of rock distributions comes primarily from three sources (1) Diviner thermophysical observations and modeling (e.g., Bandfield et al., 2011), (2) direct observation of rock abundance in Lunar Reconnaissance Orbiter Camera Narrow Angle Camera (LROC NAC) data (or earlier high resolution orbital observations) that can directly resolve boulders larger than ~2 m (e.g., Li et al., 2017), and (3) surface observations from past landing sites (e.g., Shoemaker and Morris, 1970; Muehlberger et al., 1972; Di et al., 2016).

LROC NAC data and Diviner data are available and can be used for detailed characterization of specific landing sites. The agreement of Diviner observations with direct observations of past landing sites (e.g., Li et al., 2017) suggests that it is reliable for characterizing potential hazards due to rocks. Note that Diviner rock abundance models are not computed for the poles.